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Class Activity 2

Task 1

Problem Statement

Develop a Python program to collect the details of 10 students and store them in a CSV file named student_marks.csv.
The file should contain: Register Number, Student Name, Department, Python Marks, Data Science Marks.

Python Solution Code:

```
import csv

def create_student_marks_csv():
    # Sample data for 10 students with numeric register numbers
    students = [
        {"Register Number": "192324298", "Student Name": "Alice Smith",
"Department": "Computer Science", "Python Marks": 88, "Data Science Marks": 92},
        {"Register Number": "192325346", "Student Name": "Bob Johnson",
"Department": "Information Technology", "Python Marks": 75, "Data Science Marks":
80},
        {"Register Number": "192311456", "Student Name": "Charlie Brown",
"Department": "Computer Science", "Python Marks": 95, "Data Science Marks": 89},
        {"Register Number": "192372153", "Student Name": "Diana Prince",
"Department": "Data Science", "Python Marks": 91, "Data Science Marks": 96},
        {"Register Number": "192348120", "Student Name": "Ethan Hunt", "Department":
"Cyber Security", "Python Marks": 82, "Data Science Marks": 85},
        {"Register Number": "192389045", "Student Name": "Fiona Gallagher",
"Department": "Information Technology", "Python Marks": 78, "Data Science Marks":
83},
        {"Register Number": "192356178", "Student Name": "George Clark",
"Department": "Computer Science", "Python Marks": 89, "Data Science Marks": 90},
        {"Register Number": "192334901", "Student Name": "Hannah Abbott",
"Department": "Data Science", "Python Marks": 94, "Data Science Marks": 97},
        {"Register Number": "192367823", "Student Name": "Ian Malcolm",
"Department": "Cyber Security", "Python Marks": 86, "Data Science Marks": 88},
        {"Register Number": "192312984", "Student Name": "Julia Roberts",
"Department": "Computer Science", "Python Marks": 90, "Data Science Marks": 91}
    ]
```

```
    fieldnames = ["Register Number", "Student Name", "Department", "Python Marks",
"Data Science Marks"]

    with open("student_marks.csv", mode="w", newline="", encoding="utf-8") as file:
        writer = csv.DictWriter(file, fieldnames=fieldnames)
        writer.writeheader()
        writer.writerows(students)

    print("student_marks.csv created successfully with 10 records.")

if __name__== "__main__":
    create_student_marks_csv()
```

Task 2 - CSV File: Employee Database

Problem Statement

Develop a Python program to collect employee information and store it in a CSV file named employees.csv.
The file should contain: Employee ID, Employee Name, Department, Designation, Salary.
Store details of at least 5 employees.

Python Solution Code:

```
import csv

def create_employees_csv():
    employees = [
        {"Employee ID": "EMP001", "Employee Name": "John Doe", "Department":
"Engineering", "Designation": "Senior Software Engineer", "Salary": 95000},
        {"Employee ID": "EMP002", "Employee Name": "Jane Smith", "Department":
"Human Resources", "Designation": "HR Manager", "Salary": 75000},
        {"Employee ID": "EMP003", "Employee Name": "Michael Scott", "Department":
"Sales", "Designation": "Regional Manager", "Salary": 85000},
        {"Employee ID": "EMP004", "Employee Name": "Pam Beesly", "Department":
"Administration", "Designation": "Administrator", "Salary": 55000},
        {"Employee ID": "EMP005", "Employee Name": "Jim Halpert", "Department":
"Sales", "Designation": "Assistant Regional Manager", "Salary": 78000}
    ]

    fieldnames = ["Employee ID", "Employee Name", "Department", "Designation",
"Salary"]

    with open("employees.csv", mode="w", newline="", encoding="utf-8") as file:
        writer = csv.DictWriter(file, fieldnames=fieldnames)
        writer.writeheader()
        writer.writerows(employees)

    print("employees.csv created successfully with employee details.")
```

```
if _name__== "_main_":
    create_employees_csv()
```

Task 3 - JSON File: Faculty Information System

Problem Statement

Create a Python program to collect faculty details and store them in a JSON file named faculty.json.
Each faculty record should contain: Faculty ID, Faculty Name, Department, Designation, Experience (Years).
Store details of at least 5 faculty members.

Python Solution Code:

```
import json

def create_faculty_json():
    faculty_data = [
        {"Faculty ID": "FAC001", "Faculty Name": "Dr. Alan Turing", "Department":
"Computer Science", "Designation": "Professor", "Experience (Years)": 15},
        {"Faculty ID": "FAC002", "Faculty Name": "Dr. Grace Hopper", "Department":
"Information Technology", "Designation": "Associate Professor", "Experience
(Years)": 12},
        {"Faculty ID": "FAC003", "Faculty Name": "Dr. Barbara Liskov", "Department":
"Software Engineering", "Designation": "Professor", "Experience (Years)": 18},
        {"Faculty ID": "FAC004", "Faculty Name": "Dr. Donald Knuth", "Department":
"Computer Science", "Designation": "Professor Emeritus", "Experience (Years)": 25},
        {"Faculty ID": "FAC005", "Faculty Name": "Dr. Ada Lovelace", "Department":
"Data Science", "Designation": "Assistant Professor", "Experience (Years)": 6}
    ]

    with open("faculty.json", mode="w", encoding="utf-8") as file:
        json.dump(faculty_data, file, indent=4)

    print("faculty.json created successfully with faculty details.")

if __name___== "__main__":
    create_faculty_json()
```

Task 4 - JSON File: Product Inventory Management

Problem Statement

Develop a Python program to collect product information and store it in a JSON file named products.json.

Each product should contain: Product ID, Product Name, Category, Price, Quantity Available.
Store details of at least 10 products.

Python Solution Code:

```
import json

def create_products_json():
    products = [
        {"Product ID": "P001", "Product Name": "Laptop Pro 15", "Category":
"Electronics", "Price": 1299.99, "Quantity Available": 25},
        {"Product ID": "P002", "Product Name": "Wireless Mouse", "Category":
"Accessories", "Price": 29.99, "Quantity Available": 150},
        {"Product ID": "P003", "Product Name": "Mechanical Keyboard", "Category":
"Accessories", "Price": 89.99, "Quantity Available": 80},
        {"Product ID": "P004", "Product Name": "UltraWide Monitor 34"", "Category":
"Electronics", "Price": 499.99, "Quantity Available": 30},
        {"Product ID": "P005", "Product Name": "Noise-Canceling Headphones",
"Category": "Audio", "Price": 199.99, "Quantity Available": 60},
        {"Product ID": "P006", "Product Name": "HD Webcam 1080p", "Category":
"Accessories", "Price": 59.99, "Quantity Available": 100},
        {"Product ID": "P007", "Product Name": "Ergonomic Office Chair", "Category":
"Furniture", "Price": 249.99, "Quantity Available": 20},
        {"Product ID": "P008", "Product Name": "USB-C Docking Station", "Category":
"Electronics", "Price": 119.99, "Quantity Available": 45},
        {"Product ID": "P009", "Product Name": "External SSD 1TB", "Category":
"Storage", "Price": 99.99, "Quantity Available": 90},
        {"Product ID": "P010", "Product Name": "Smart LED Desk Lamp", "Category":
"Furniture", "Price": 39.99, "Quantity Available": 70}
    ]

    with open("products.json", mode="w", encoding="utf-8") as file:
        json.dump(products, file, indent=4)

    print("products.json created successfully with 10 product items.")

if __name___== "__main__":
    create_products_json()
```

Task 5-XML File: Course Information Database

Problem Statement

Create a Python program to collect course details and store them in an XML file named courses.xml.
Each course should contain: Course Code, Course Name, Credits, Department, Faculty Name.
Store details of at least 5 courses.

Python Solution Code:

```
import xml.etree.ElementTree as ET
from xml.dom import minidom

def create_courses_xml():
    courses = [
        {"Course Code": "CS101", "Course Name": "Python Programming", "Credits":
"4", "Department": "Computer Science", "Faculty Name": "Dr. Alan Turing"},
        {"Course Code": "DS201", "Course Name": "Data Science Fundamentals",
"Credits": "3", "Department": "Data Science", "Faculty Name": "Dr. Grace Hopper"},
        {"Course Code": "SE301", "Course Name": "Software Architecture", "Credits":
"4", "Department": "Software Engineering", "Faculty Name": "Dr. Barbara Liskov"},
        {"Course Code": "CS401", "Course Name": "Advanced Algorithms", "Credits":
"4", "Department": "Computer Science", "Faculty Name": "Dr. Donald Knuth"},
        {"Course Code": "AI101", "Course Name": "Introduction to AI", "Credits":
"3", "Department": "Artificial Intelligence", "Faculty Name": "Dr. Ada Lovelace"}
    ]

    root = ET.Element("Courses")

    for c in courses:
        course_elem = ET.SubElement(root, "Course")
        for key, val in c.items():
            tag_name = key.replace(" ", "")
            child = ET.SubElement(course_elem, tag_name)
            child.text = str(val)

    xml_str = minidom.parseString(ET.tostring(root, encoding="utf-
8")).toprettyxml(indent="  ")

    with open("courses.xml", mode="w", encoding="utf-8") as file:
        file.write(xml_str)

    print("courses.xml created successfully with course details.")

if __name___== "__main__":
    create_courses_xml()
```

Task 6 - XML File: Hospital Patient Management System

Problem Statement

Develop a Python program to collect patient information and store it in an XML file named patients.xml.
Each patient record should contain: Patient ID, Patient Name, Age, Disease, Doctor Assigned.
Store details of at least 10 patients.

Python Solution Code:

```
import xml.etree.ElementTree as ET
from xml.dom import minidom

def create_patients_xml():
    patients = [
        {"Patient ID": "P101", "Patient Name": "Arthur Dent", "Age": "42",
"Disease": "Hypertension", "Doctor Assigned": "Dr. House"},
        {"Patient ID": "P102", "Patient Name": "Sarah Connor", "Age": "35",
"Disease": "Fracture", "Doctor Assigned": "Dr. Watson"},
        {"Patient ID": "P103", "Patient Name": "Bruce Wayne", "Age": "40",
"Disease": "Concussion", "Doctor Assigned": "Dr. Strange"},
        {"Patient ID": "P104", "Patient Name": "Clark Kent", "Age": "33", "Disease":
"Kryptonite Allergy", "Doctor Assigned": "Dr. McCoy"},
        {"Patient ID": "P105", "Patient Name": "Diana Prince", "Age": "29",
"Disease": "Influenza", "Doctor Assigned": "Dr. House"},
        {"Patient ID": "P106", "Patient Name": "Peter Parker", "Age": "22",
"Disease": "Insect Bite", "Doctor Assigned": "Dr. Watson"},
        {"Patient ID": "P107", "Patient Name": "Tony Stark", "Age": "48", "Disease":
"Arrhythmia", "Doctor Assigned": "Dr. Strange"},
        {"Patient ID": "P108", "Patient Name": "Natasha Romanoff", "Age": "34",
"Disease": "Laceration", "Doctor Assigned": "Dr. McCoy"},
        {"Patient ID": "P109", "Patient Name": "Steve Rogers", "Age": "101",
"Disease": "Fatigue", "Doctor Assigned": "Dr. House"},
        {"Patient ID": "P110", "Patient Name": "Wanda Maximoff", "Age": "28",
"Disease": "Migraine", "Doctor Assigned": "Dr. Strange"}
    ]

    root = ET.Element("Patients")

    for p in patients:
        patient_elem = ET.SubElement(root, "Patient")
        for key, val in p.items():
            tag_name = key.replace(" ", "")
            child = ET.SubElement(patient_elem, tag_name)
            child.text = str(val)

    xml_str = minidom.parseString(ET.tostring(root, encoding="utf-
8")).toprettyxml(indent="  ")

    with open("patients.xml", mode="w", encoding="utf-8") as file:
        file.write(xml_str)

    print("patients.xml created successfully with 10 patient records.")

if __name___== "__main__":
    create_patients_xml()
```